Agentic AI for IP Network Operation (AgenticOPS)

IETF 125 Shenzhen Meeting

March 16 2026, Tuesday 16:30~18:00

ZhenQiang Li, Daniel King, Qin Wu as Organizers

Agenda: https://github.com/danielkinguk/AgenticOps/tree/main

Material:https://github.com/danielkinguk/AgenticOps/tree/main/materials

Note Well

This is a reminder of IETF policies in effect on various topics such as patents or code of conduct. It is only meant to point you in the right direction. Exceptions may apply. The IETF's patent policy and the definition of an IETF "contribution" and "participation" are set forth in BCP 79; please read it carefully.

As a reminder:

- By participating in the IETF, you agree to follow IETF processes and policies.
- If you are aware that any IETF contribution is covered by patents or patent applications that are owned or controlled by you or your sponsor, you must disclose that fact, or not participate in the discussion.
- As a participant in or attendee to any IETF activity you acknowledge that written, audio, video, and photographic records of meetings may be made public.
- Personal information that you provide to IETF will be handled in accordance with the IETF Privacy Statement.
- As a participant or attendee, you agree to work respectfully with other participants; please contact the ombudsteam if you have questions or concerns about this.

Definitive information is in BCP 9, BCP 25, BCP 54, BCP 78, BCP 79, and the IETF Privacy Policy.
For advice, please talk to WG chairs or ADs.

Motivation and Goal

- Background:
  - Network operators, vendors of network equipment, and implementers of NMS/OSS have Launched many Innovation projects related to Multi-Agent Collaboration across layer, Domain and Vendors
  - TMF has recently published A2A-T for Telecom graded Network Operation.
  - Agentic AI pose great challenges to Network management architecture
    - Security
    - Protocol

Goal:

- Explore applicability, architecture design, and candidate interfaces and protocols related to AgenticOPS

- Share information and determine what future work might belong in the IETF

Agenda for Today's Side Meeting

Time	Items
5 minutes	Opening remarks — IETF meeting rules, agenda arrangement, and objectives
75 minutes	Unexpected Incidents with Multi-Agent Collaboration (Roland Schott & Nils.Warnke - Deutsche Telekom)
	Network Anomaly Detection with Agentic AI (Thomas Graf &Wanting Du-Swisscom)
	Network Changes enabled by Agent and Network Digital Twin (Ying Si - China Mobile)
	Agentic AI for IP Network Operations (Jiangtao LI&Hongyang Liu - China Unitechs)
	Agentic AI Driven Network Management Migration Challenges and Opportunities in IETF (Qin Wu - Huawei)
	AI based Network Management Agent: Industry Direction and Standards Opportunities (Daniele Ceccarelli & Xing James Jiang - Cisco)
10 minutes	Closing summary and next steps

Open Discussion

1. How are network management architecture and management interface evolved with Agentic AI.
2. Do we need to have common terminologies and Use cases for Agentic AI Network Operation?
3. What work might there be for the IETF?

Potential IETF work items

- Common Agentic AI Network Management Architecture for Human to Agent, Agent to Tools, Agent to Agent Interaction
- Common Terminology and Use Cases for AI Network Management Operation
- Cross Layer/Cross Domain Agent to Agent management interface for high risk, reliability, real time network operation.
- Agent prompt language templates for IETF intent
- Human-Agent Collaboration Interfaces for control,observability and explainable capability

Common Use cases: Fault Management, Network Optimization for Service Assurance, Network Change Management

Backup slides

AI for Network Operation Coordination

Network Operation
Agentic ai architecture and PS
MCP/A2A Applicability
Agentic Network Inference Architecture
AI based Network Management Architecture

LLM/AI Benchmarking
Terminology
Benchmarking
Profile

Inter-Agent
Natural Language for Agent Communication
Aiprotocol agent to tools
Aiprotocol framework
Aiprotocol cheq

Discovery and Naming
Selective Disclosure for Agent Discovery
DNS-Native AI Agent Naming and Resolution
Agent Directory Service

Security
AI Agent Authentication and Authorization
Delegated Agent Authorization Protocol
Agent Operation Authorization

Transport
Agent Protocol over MoQ
MCP over MoQ
Agent Communication over LISP
Agntcy Messaging

- AI for Network Operation Repo: https://github.com/IETF-OPS-AD/AINETOPS

AI for Network Operation Coordination

- Motivation:
  - AI activity is a jungle, Side Meetings are dominated by AI
    - Each group move independently in the side meeting
    - Too much unstructured work, too many topics, people are lost.
  - CATALIST BoF is a good start in trying to identify the what the IETF needs to be working on, but not specific to Network Management

- Goal:
  - Use this side meeting coordinating discussion and highlighting work related to network management, flesh out concrete pieces which can be done in formal WG
  - Continue to use ainetops@ietf.org mailing list to bridge various different work and side meetings

Discussion List: https://mailarchive.ietf.org/arch/browse/ainetops/

AI for Network Operation Work Is Everywhere

Topic	Draft Name and Title	WG/RG	Status
Architecture	draft-wmz-nmrg-agent-ndt-arch-03 Network Digital Twin and Agentic AI based Architecture for AI driven Network Operations	NMRG	Individual draft
Problem Statement	draft-hong-nmrg-agenticai-ps-01 Motivations and Problem Statement of Agentic AI for network management	NMRG	Individual draft
Concept	draft-jadoon-nmrg-agentic-ai-autonomous-networks-00 Agentic AI Architectural Principles for Autonomous Computer Networks	NMRG	Individual draft
Architecture	draft-zhao-nmrg-ai-agent-for-ndt-00 AI Agent Architecture for Network Digital Twin	NMRG	Individual draft
Architecture	draft-chuyi-nmrg-agentic-network-inference-00 Agentic Network Architecture and Protocol for Supporting Agent Interconnection Communication and Multi-level Inference	NMRG	Individual draft
Framework	draft-zhao-ccamp-actn-optical-network-agent-01 Integration of Network Management Agent (NMA) into ACTN-Based Optical Network	CCAMP	Individual draft
Architecture	draft-zhao-nmop-network-management-agent-04 AI based Network Management Agent(NMA): Concepts and Architecture	NMOP	Individual draft
Framework	draft-fu-nmop-agent-communication-framework-00 Agent Communication Framework for Network AIOps	NMOP	Individual draft
Use Case and Requirement	draft-han-anima-ai-asa-01 Considerations of AI-powered Autonomic Service Agent Communication	ANIMA	Individual draft
Architecture	draft-ietf-nmop-network-anomaly-architecture-07 A Framework for a Network Anomaly Detection Architecture	NMOP	WG draft
YANG Model	draft-ietf-nmop-network-anomaly-semantics-05 Semantic Metadata Annotation for Network Anomaly Detection	NMOP	WG draft
YANG Model	draft-ietf-nmop-network-anomaly-lifecycle-05 An Experiment: Network Anomaly Detection Lifecycle	NMOP	WG draft
YANG Model	draft-ietf-nmop-network-incident-yang-08 A YANG Data Model for Network Incident Management	NMOP	WG draft

AI for Network Operation Related Work is Everywhere

Topic	Draft Name and Title	WG/RG	Status
Benchmark	draft-gaikwad-llm-benchmarking-terminology-00 Benchmarking Terminology for Large Language Model Serving	BMWG	Individual draft
Benchmark	draft-gaikwad-llm-benchmarking-profiles-00 Performance Benchmarking Profiles for Large Language Model Serving Systems	BMWG	Individual draft
Benchmark	draft-gaikwad-llm-benchmarking-methodology-00 Benchmarking Methodology for Large Language Model Serving	BMWG	Individual draft
Benchmark	draft-calabria-bmwg-ai-fabric-terminology-00 Benchmarking Terminology for AI Network Fabrics	BMWG	Individual draft
Benchmark	draft-calabria-bmwg-ai-fabric-inference-bench-00 Benchmarking Methodology for AI Inference Serving Network Fabrics	BMWG	Individual draft
Benchmark	draft-calabria-bmwg-ai-fabric-training-bench-00 Benchmarking Methodology for AI Training Network Fabrics	BMWG	Individual draft
Benchmark	draft-cui-nmrg-auto-test-01 Framework and Automation Levels for AI-Assisted Network Protocol Testing	NMRG	Individual draft
Use Case	draft-zeng-nmrg-mcp-usecases-requirements-00 MCP for Network Management: Problem Statement, Use Cases, and Requirements	NMRG	Individual draft
Use case	draft-irtf-nmrg-ibn-usecases-02 Use Cases and Practices for Intent-Based Networking	NMRG	RG Draft
Use Case	draft-bernardos-nmrg-agentic-network-optimization-00 Solutions for enabling agentic sensing with network optimization	NMRG	Individual draft
Framework	draft-irtf-nmrg-ai-deploy-02 Considerations of network/system for AI services	NMRG	RG Draft
Protocol	draft-yang-nmrg-mcp-nm-02 Applicability of MCP for the Network Management	NMRG	Individual draft
Protocol	draft-yang-nmrg-a2a-nm-02 Applicability of A2A to the Network Management	NMRG	Individual draft
Protocol	draft-verma-dmsc-nlip-notes-00 Use of Natural Language for Agent Communication	DMSC Group	Individual draft